Highly accurate protein structure prediction with AlphaFold

Paper by Jumper et al., *Nature* 596 (2021)

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Machine Learning for Genomics (263-5351-00L), ETH Zürich, Fall 2026

The problem: sequence to structure

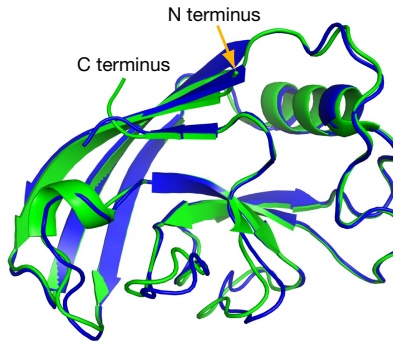
`SIFSYITESTGTPS...` →

Primary amino acid sequence $\mathbf{s} \in \mathcal{A}^L$

Genomic sequencing yields $> 2 \times 10^8$ known proteins (UniProt); structural experiments cover $< 2 \times 10^5$ (PDB).

Thermodynamic fold determination (Anfinsen) faces astronomical conformational search (10^{143} states, Levinthal).

Predict all heavy-atom 3D coordinates $\mathbf{x} \in \mathbb{R}^{N_{\text{atoms}} \times 3}$ directly from primary sequence.



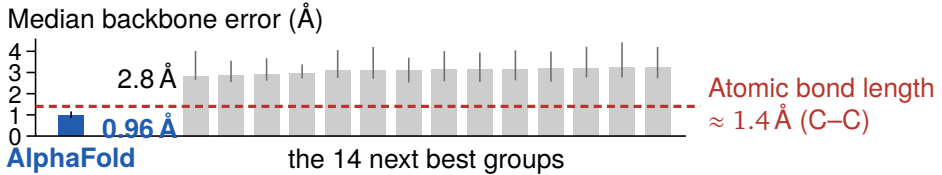
Ground truth in green (PDB 6Y4F), prediction in blue (0.8 Å RMSD)

[JEP+21] Jumper et al., *Nature* 2021, Fig. 1b. Target T1049 (PDB 6Y4F), CASP14.

The benchmark: CASP14

CASP14 (May–July 2020): double-blind evaluation on experimentally determined, unreleased structures.

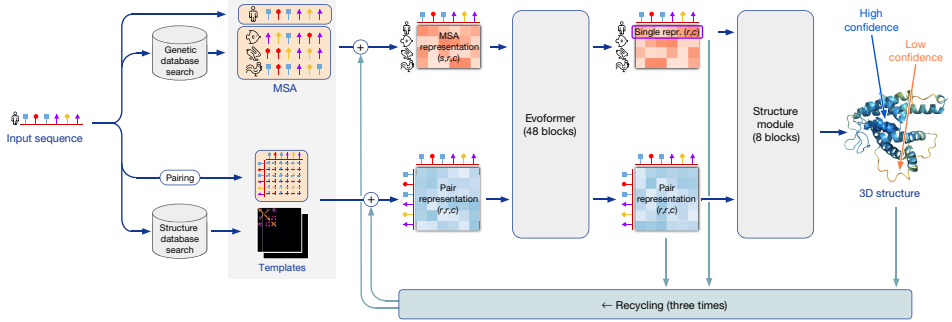
Evaluation metric: Cα root-mean-square deviation on 95% best-model residues (r.m.s.d.$_{95}$).



Median backbone error reaches 0.96 Å—surpassing typical covalent C–C bond length (1.4 Å).

Redrawn from Fig. 1a. Cα r.m.s.d.$_{95}$ on 87 domains (95% CIs). 1 Å = 10^{-10} m.

The pipeline: MSA in, coordinates out



End-to-end differentiable pipeline: unifies representation learning and coordinate generation without intermediate hand-crafted potentials.

[JEP+21] Jumper et al., *Nature* 2021, Fig. 1e (Evoformer: 48 blocks; Structure Module: 8 iterations).

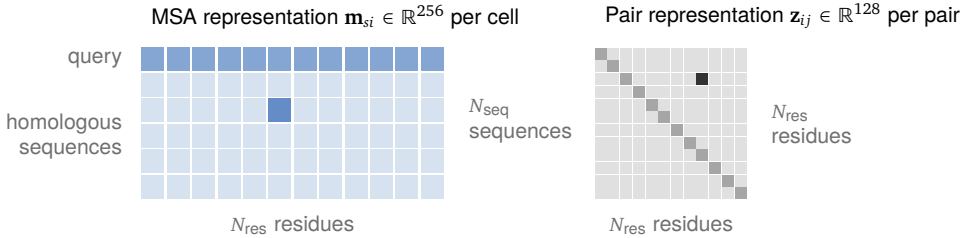
Four facts, four design choices

		Fact	Design choice in AlphaFold 2
1		MSA is a set; chain is 1D	Axial attention on rows and columns
2		3D distances close triangles	Triangle updates on residue pair graph
3		Rigid motion invariance: SE(3)	Local frames; Invariant Point Attention
4		Proteins are chiral; mirrors differ	Frame Aligned Point Error (FAPE) loss

Removing any one of the four costs accuracy: the ablation study measures each.

The two representations

Dual-track representations: MSA tensor $\mathbf{m}_{si}$ and residue pair matrix $\mathbf{z}_{ij}$

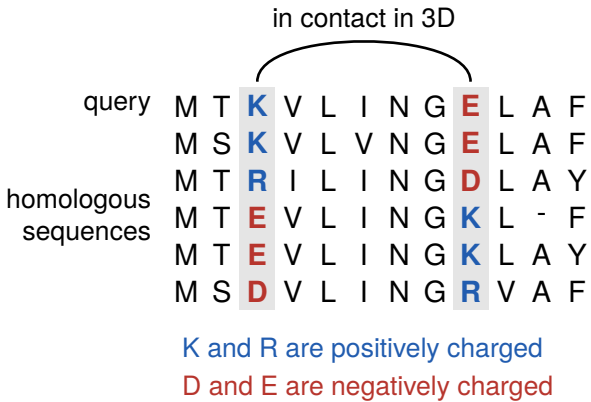


The MSA tensor maintains sequence-position states ($c_m = 256$); the pair matrix maintains residue-residue relational states ($c_z = 128$).

The 48 Evoformer blocks iteratively refine both representations via bidirectional cross-talk.

Supplementary Information (SI) 1.5, Algorithms 3 and 4. Sizes in training: 128 sequences, crops of 256 residues (SI Table 4).

The input: MSA representation



Rows index homologous evolutionary sequences; columns index aligned residue positions along the query chain.

Compensatory co-mutations (e.g. salt-bridge charge conservation) reflect direct 3D physical contacts.

AlphaFold processes the raw alignment directly, learning epistatic coupling patterns end-to-end.

[MHC+11] D. S. Marks et al., *PLoS ONE* 6, 2011. The alignment is an illustration, not data.

The input: Pair representation

Relational graph initialization and spatial priors

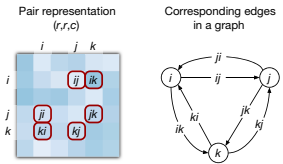


Fig. 3b, the pairs as directed edges

Residue pairs (i, j) form directed edges in a complete residue graph ($\mathbf{z}_{ij} \in \mathbb{R}^{128}$).

Initialized from relative sequence separation $|i - j|$ and homologous structural template distances.

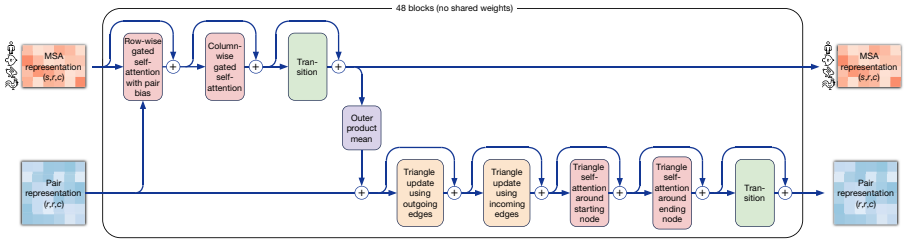
Supervised auxiliary head: a 64-bin distogram predicts C_β–C_β distance distributions directly from $\mathbf{z}_{ij}$.

The pair representation maintains spatial relations before 3D coordinates are instantiated.

SI 1.5, Algorithm 4 (relative position), 1.7.1 (templates), 1.10 (recycling), 1.9.8 (distogram).

The Evoformer block

1. The sequences in an MSA are a set. The residues are a chain.



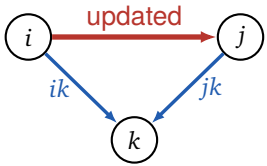
Axial attention alternates over residue positions (rows) and homologous sequences (columns).

Bidirectional coupling: pair edges bias row attention; outer product mean injects evolutionary covariance into edge features.

[JEP+21] Jumper et al., *Nature* 2021, Fig. 3a (48 blocks, SI Algorithms 7–10). Full update equations in backup 11–12.

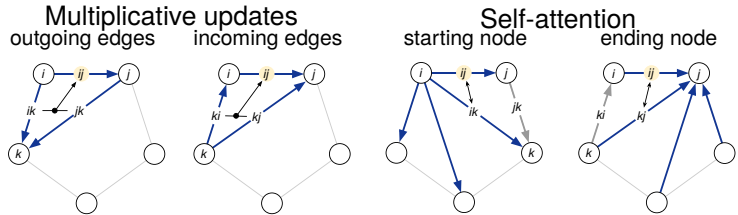
Evoformer: triangle updates

2. Distances in 3D must close triangles.



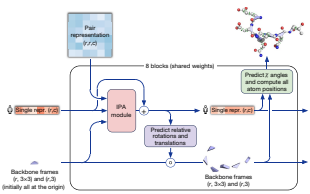
Edge (i, j) is updated by aggregating paths through intermediate residues k: $\sum_k \mathbf{z}_{ik} \odot \mathbf{z}_{jk}$.

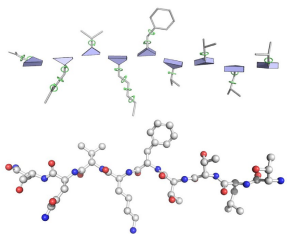
Triangle updates enforce metric spatial constraints and the triangle inequality directly in 2D pair space.



Structure module: residues as rigid bodies

3. A structure is the same after any rotation or translation.





Each residue is parameterized by an SE(3) rigid-body frame $T_i = (R_i, \vec{\mathbf{t}}_i)$; torsion angles χ place sidechains.

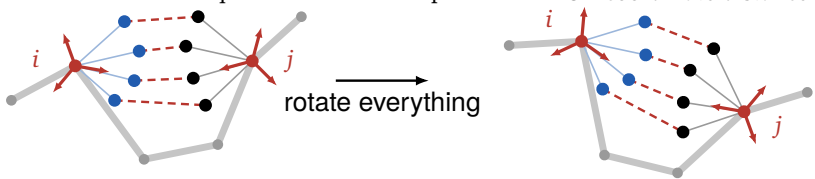
Residues update simultaneously in local coordinates as an unconstrained "residue gas"; peptide bond stereochemistry is learned.

[JEP+21] Jumper et al., *Nature* 2021, Fig. 3c,d (8 iterations, SI Algorithms 20–23).

Invariant Point Attention (IPA)

3. A structure is the same after any rotation or translation.

$$a_{ij} = \text{softmax}_j\Big(\underbrace{\boldsymbol{q}_i^\top \boldsymbol{k}_j}_{\text{scalar sequence attention}} + \underbrace{b_{ij}}_{\text{pair bias}} - \underbrace{\textstyle\sum_p \left\| T_i \circ \vec{\boldsymbol{q}}_{ip} - T_j \circ \vec{\boldsymbol{k}}_{jp} \right\|^2}_{\text{3D coordinate distance penalty}} \Big)$$



Points are placed in the frames of i and j

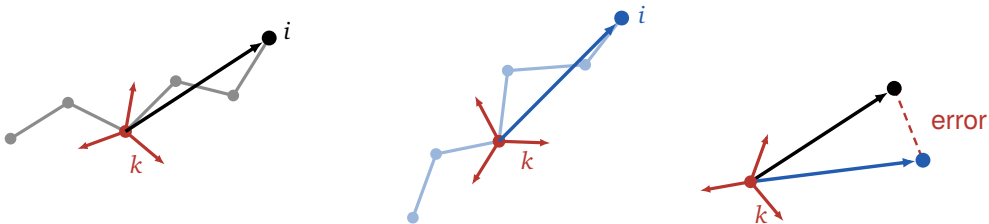
The distances stay the same

Exact algebraic $\mathrm{SE}(3)$-invariance: attention weights and scalar updates are independent of global frame.

SI 1.8.2, Algorithm 22 (single-head formulation). Algebraic invariance proof in backup 1.

Frame Aligned Point Error (FAPE)

4. A protein and its mirror image are different molecules.



ground truth | prediction | both, seen from frame k

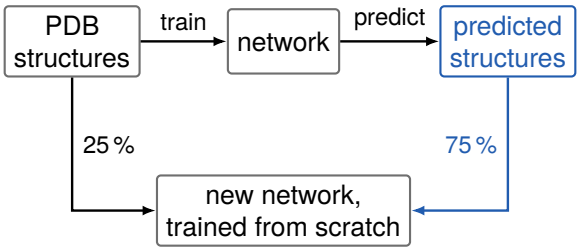
$$\text{FAPE} = \frac{1}{Z} \underset{k,i}{\text{mean}} \min\left(d_{\text{clamp}}, \left\| (\boldsymbol{R}_k, \boldsymbol{t}_k)^{-1} \boldsymbol{x}_i - (\boldsymbol{R}_k^{\text{true}}, \boldsymbol{t}_k^{\text{true}})^{-1} \boldsymbol{x}_i^{\text{true}} \right\|\right)$$

Local-frame point alignment: rigorously SE(3)-invariant, strictly sensitive to chiral enantiomer reflections.

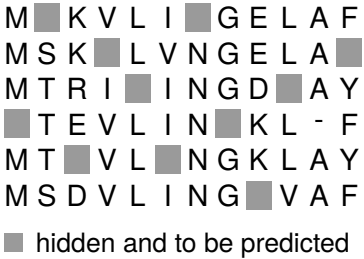
Supplementary Information, Algorithm 28, with $Z = d_{\text{clamp}} = 10$ Å. Pseudometric proof in backup 3–4.

Training data: PDB and self-distillation

Self-distillation on 350,000 unlabelled sequences triples data scale (75% synthetic batches).



Masked-MSA loss (15% masked amino acids) trains Evoformer representations without structural labels.

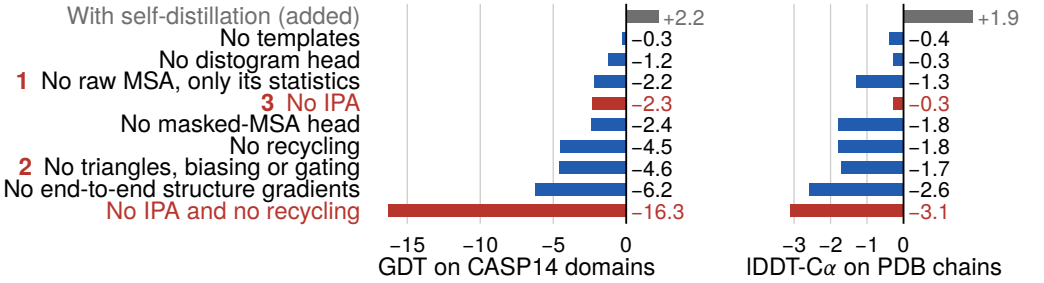


Dual self-supervision: leverages uncharacterized sequence databases to generalize beyond PDB coverage.

[XLHL20] Xie et al., CVPR 2020 (self-distillation). [DCLT19] Devlin et al., NAACL 2019 (BERT masking).

Ablations: the cost of each component

1 2 3 4

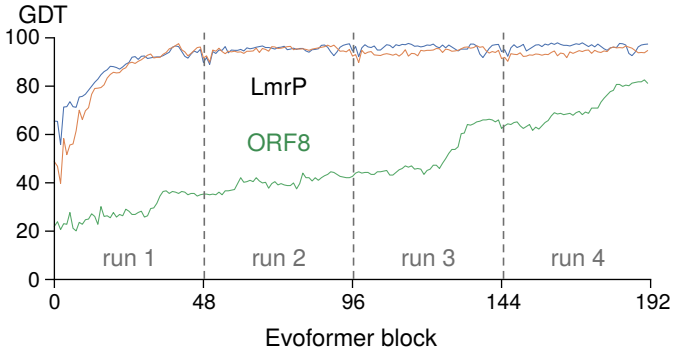


4 Without FAPE stereochemical grounding, reflection-invariant losses produce 50% inverted chiral enantiomers (Suppl. Fig. 9).

Severe non-linear coupling: removing both IPA and recycling causes a catastrophic −16.3 GDT collapse.

Redrawn from Fig. 4a, mean change vs baseline over training runs. GDT and lDDT 0–100. Row 3 (No IPA) removes iterative refinement but keeps frames and FAPE (SI 1.13.1). Line 4: Suppl. Fig. 9.

Intermediate structures in the trunk

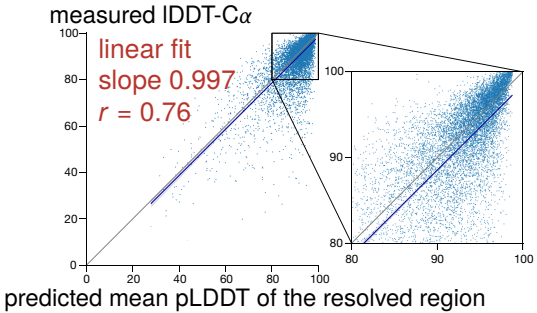


Structures can be probed at intermediate Evoformer blocks by applying a shared Structure Module readout.

Bimodal folding dynamics: compact domains (LmrP) converge in early blocks; complex topologies (ORF8) require full depth.

[JEP+21] Jumper et al., *Nature* 2021, Fig. 4b. LmrP: transporter (PDB 6T1Z); ORF8: SARS-CoV-2 (T1064).

Confidence: pLDDT against measured lDDT



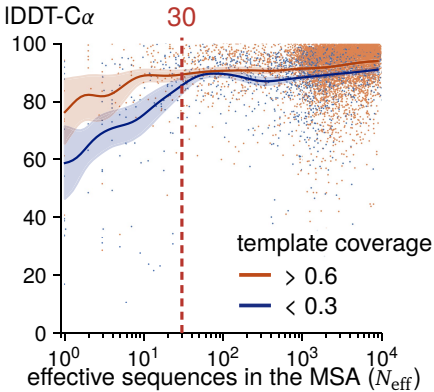
pLDDT predicts local lDDT-Cα accuracy (r = 0.76, slope 0.997 vs measured truth).

Low scores (pLDDT < 50) accurately flag intrinsically disordered regions (IDRs).

Calibrated uncertainty: reliable residue-level error estimation without experimental ground truth.

[JEP+21] Jumper et al., *Nature* 2021, Fig. 2c, n = 10,795 chains. Following [TAW+21] Tunyasuvunakool et al., *Nature* 2021.

Limitations: MSA depth and complexes



Prediction fidelity degrades sharply below an evolutionary depth threshold of $N_{\text{eff}} \approx 30$ (unless rescued by templates).

Monomer models fail on obligate multimers whose tertiary folds are stabilized exclusively by quaternary interfaces.

AlphaFold 2 predicts static, isolated monomeric ground states, omitting allosteric conformational ensembles.

[JEP+21] Jumper et al., *Nature* 2021, Fig. 5a.

After AlphaFold 2: 2021 to 2025

2021 RoseTTAFold published simultaneously [BDI+21]

98.5% of the human proteome predicted [TAW+21]

AlphaFold-Multimer: multimeric complexes [EOP+21]

2022 ESMFold: language model, single sequence [LAR+23]

2024 AlphaFold 3: all biomolecular complexes [AAD+24]

directly address the limits

Nobel Prize in Chemistry (Baker, Hassabis, Jumper, 2024)

2025 AlphaGenome: variant effects in non-coding DNA [Avs+25]

Expansion of structural genomics: from monomeric static predictions to proteome-scale databases and all-atom complexes.

Exception: AIchemy_RNA2 on CASP15 RNA targets. Complete citations on closing References slides.

AlphaFold 3: which choices survived

Abramson et al. [AAD+24], *Nature* 630 (2024), pages 493 to 498

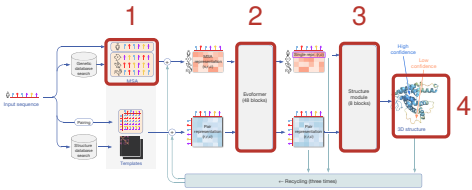
	AlphaFold 2	AlphaFold 3 verdict	The cost / patch
1	raw MSA attention	small MSA block, not kept	same MSA depth dependence
2	triangles on the pairs	kept unchanged, 48 blocks	inductive bias remains key
3	frames, IPA, equivariance	dropped for diffusion	clashes; ranking and distillation
4	FAPE in local frames	dropped; aligned MSE loss	4.4 % chirality errors; ranking, seeds

Unlike AlphaFold 2, AlphaFold 3 provides no component ablation study.

Generative paradigm shift: relational pair graphs survived; rigid kinematics yielded to all-atom diffusion.

[Sut19] R. Sutton, *The Bitter Lesson*, 2019. AF3 reports 4.4% chirality violations and relies on multi-seed ranking. Architecture: backup 14.

Summary: four design choices, measured



		GDT lost
1	attention over sequences	−2.2
2	triangles on the pairs	−4.6
3	frames and IPA	−2.3
4	FAPE in local frames	mirrored

End-to-end gradient supervision: inference executes in $\sim$ 1 GPU-minute for standard domains ($L \approx 300$).

Structural biology constraints as deep learning inductive biases: each choice quantitatively drives atomic accuracy.

Exam question

Machine Learning for Genomics (263-5351-00L)

Official Assignment

Assigned by Prof. Valentina Boeva and course instructors 7 days prior to presentation date.

[Instructor question prompt inserted here]

Structured Solution

1. Inductive bias and architecture
2. Mathematical loss formulation
3. Genomic & biological impact

Rigorous theoretical justification and step-by-step derivation

Questions for the audience

1 AlphaFold 2 hardcoded SE(3) equivariance and stereochemical chirality into architecture and loss. AlphaFold 3 replaced both with unconstrained diffusion. When is geometric inductive bias indispensable, and when is unconstrained generative modeling preferable?

2 Accuracy falls below about 30 related sequences. Has the network learned how proteins fold, or how evolution has varied them? What result would settle it?

3 pLDDT is trained on the model's own errors. On which proteins would you distrust it, and what should a genomics pipeline do with a low-confidence region?

AlphaFold Resources & Discussion: {ebosch, bben}@ethz.ch

Code, derivations & slides: github.com/BenBullinger/ML4Genomics-Presentation-AF2

References: AlphaFold & Structural Biology

Primary literature and benchmark publications

[JEP+21] J. Jumper, R. Evans, A. Pritzel, et al., “Highly accurate protein structure prediction with AlphaFold,” *Nature* 596:583–589, 2021.

[TAW+21] K. Tunyasuvunakool, P. Adler, Z. Wu, et al., “Highly accurate protein structure prediction for the human proteome,” *Nature* 596:590–596, 2021.

[EOP+21] R. Evans, M. O’Neill, A. Pritzel, et al., “Protein complex prediction with AlphaFold-Multimer,” *bioRxiv*:2021.10.04.463034, 2021.

[BDI+21] M. Baek, F. DiMaio, I. Anishchenko, et al., “Accurate prediction of protein structures and interactions using a three-track neural network,” *Science* 373:871–876, 2021.

[AAD+24] J. Abramson, J. Adler, J. Dunger, et al., “Accurate structure prediction of biomolecular interactions with AlphaFold 3,” *Nature* 630:493–500, 2024.

[MHC+11] D. S. Marks, L. J. Colwell, R. Sheridan, et al., “Protein 3D structure computed from evolutionary sequence variation,” *PLoS ONE* 6:e28766, 2011.

References: Machine Learning & Genomics

Language models, diffusion, regulatory genomics, and paradigms

[LAR+23] Z. Lin, H. Akin, R. Rao, et al., "Evolutionary-scale prediction of atomic-level protein structure with a language model," *Science* 379:1123–1130, 2023.

[Avs+25] Ž. Avsec, et al., "Predicting variant regulatory impact in non-coding DNA with AlphaGenome," *bioRxiv*, 2025.

[DCLT19] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," *NAACL-HLT*, 2019.

[XLHL20] Q. Xie, M.-T. Luong, E. Hovy, Q. V. Le, "Self-training with noisy student improves ImageNet classification," *CVPR*, 2020.

[Sut19] R. Sutton, "The Bitter Lesson," *Incomplete Ideas* (essay), 2019.

Resources: AlphaFold at ETH

Pipelines, cluster execution, and seminar contacts

Tooling & Deployment

- ✓ **ColabFold**: fast MMseqs2 monomer and multimer prediction
- ✓ **Euler Cluster**: GPU batch jobs via SLURM (`-gpus=1`)
- ✓ **AFDB**: 214M+ structures at `alphafold.ebi.ac.uk`

Repository & Contacts

- ✓ **Deck & TikZ Sources**: `github.com/BenBullinger/...`
- ✓ **Questions & Discussion**: `{ebosch, bben}@ethz.ch`
- ✓ **Seminar**: ML for Genomics (Boeva Lab)

Open-source code, derivations, and structural biology pipelines available on GitHub

Backup (1/14): Why is IPA invariant to rigid motions?

SI 1.8.2, Algorithm 22, equations 3 to 6



The only geometry in the weights is the distance between points that i and j place in their own frames

$$\left\|(T_{\text{global}} \circ T_i) \circ \vec{\boldsymbol{q}}_i - (T_{\text{global}} \circ T_j) \circ \vec{\boldsymbol{k}}_j\right\|^2$$
$$= \left\|T_{\text{global}} \circ (T_i \circ \vec{\boldsymbol{q}}_i - T_j \circ \vec{\boldsymbol{k}}_j)\right\|^2 = \left\|T_i \circ \vec{\boldsymbol{q}}_i - T_j \circ \vec{\boldsymbol{k}}_j\right\|^2$$

moving both points together leaves their distance unchanged

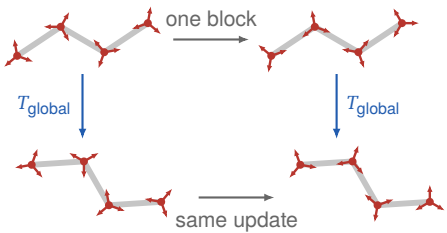
$$(T_{\text{global}} \circ T_i)^{-1} \circ \textstyle\sum_j a_{ij}\, (T_{\text{global}} \circ T_j) \circ \vec{\boldsymbol{v}}_j$$
$$= T_i^{-1} \circ T_{\text{global}}^{-1} \circ T_{\text{global}} \circ \textstyle\sum_j a_{ij}\, T_j \circ \vec{\boldsymbol{v}}_j = T_i^{-1} \circ \textstyle\sum_j a_{ij}\, T_j \circ \vec{\boldsymbol{v}}_j$$

the output points are read back in frame i, and T_{global} meets its inverse. Exact, not learned.

Weights and outputs do not depend on where the structure sits

Backup (2/14): Why is the structure module equivariant?

SI 1.8.2 and 1.8.3, Algorithms 20 and 23



Frame update predicted from invariant $\boldsymbol{s}_i$, composed in local coordinates:

$$(T_{\text{global}} \circ T_i) \circ T_i^{\text{upd}} = T_{\text{global}} \circ (T_i \circ T_i^{\text{upd}})$$

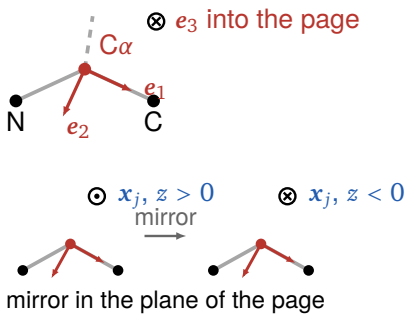
Moving input first or output last yields identical frames.

Frames initialize at identity; loss is invariant to initial pose.

Exact SE(3) equivariance. Chirality is enforced by FAPE and chiral PDB data.

Backup (3/14): Origin of stereochemical chirality

SI 1.8.1, 1.9.3 and 1.8.4, Algorithm 21, equations 11 to 17



Residue frames from (N, Cα, C) via Gram–Schmidt ($\boldsymbol{e}_3 = \boldsymbol{e}_1 \times \boldsymbol{e}_2$): always right-handed.

Reflecting all atoms leaves frames right-handed; only local z flips:

$$T_i^{-1} \circ \boldsymbol{x}_j \;\mapsto\; \mathrm{diag}(1, 1, -1)\,(T_i^{-1} \circ \boldsymbol{x}_j)$$

$$\mathrm{FAPE}(X, X_{\mathrm{mirror}}) = 2 \sum_{ij} \left|(T_i^{-1} \circ \boldsymbol{x}_j)_z\right|$$

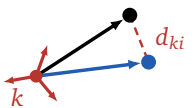
Non-zero unless structure is strictly planar.

Models trained with reflection-blind dRMSD produce mirror images frequently (Suppl. Fig. 9).

FAPE breaks reflection symmetry; PDB training data teach L-amino acid chirality.

Backup (4/14): What does FAPE measure?

SI 1.9.2, 1.9.4 and 1.9.5, Algorithm 28, equations 18 to 30



seen from frame k

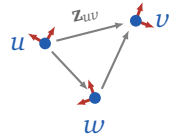
- **Local coordinates**: Evaluates all $N_{\text{frames}} \times N_{\text{atoms}}$ pairs in frame k, clamped at $d_{\text{clamp}} = 10\,\text{Å}$.
- **Rigid motion invariance**: Invariant under separate SE(3) rotations and translations of prediction and target.
- **Optimal alignment lower bound**: FAPE $\geq$ RMSD after optimal superposition; zero if and only if poses agree up to rigid motion.
- **Mathematical metric**: Non-negative, symmetric, and satisfies the triangle inequality (pseudometric on structures).

Local geometry in every frame: blind to placement, sensitive to the hand

Backup (5/14): Is AlphaFold 2 a graph neural network?

Formalism of Bronstein, Bruna, Cohen and Veličković, 2021

$$\mathbf{h}_u = \phi\Big(\mathbf{x}_u, \bigoplus_{v \in \mathcal{N}_u} \psi(\mathbf{x}_u, \mathbf{x}_v, \mathbf{e}_{uv})\Big)$$



message ψ, aggregation $\bigoplus$, update ϕ. Nodes: residues, $\mathbf{s}_u$ and frame T_u. Edges: $\mathbf{z}_{uv}$.

Block	message ψ	into
MSA row attention	$\psi(\mathbf{m}_{su}, \mathbf{m}_{sv}, \mathbf{z}_{uv})$, the edge biases the weight	node u
outer product mean	$\psi(\mathbf{m}_{\cdot u}, \mathbf{m}_{\cdot v})$, averaged over the sequences	edge uv
triangle updates	$\psi(\mathbf{z}_{uv}, \mathbf{z}_{uw}, \mathbf{z}_{vw})$ over third nodes w (out and in)	edge uv
IPA	$\psi(\mathbf{s}_u, \mathbf{s}_v, \mathbf{z}_{uv}, \lVert T_u \circ \vec{\mathbf{q}}_u - T_v \circ \vec{\mathbf{k}}_v \rVert)$, values moved by $T_u^{-1} \circ T_v$	node u
backbone update	none, ϕ alone	frame T_u

Triangles combine two edges into the third to enforce 3D spatial consistency.

Geometry enters through invariants: distances in weights, relative frames in values.

Backup (6/14): What does each block compute?

Supplement, Algorithms 6 to 32

	Update	Message	SI
	Evoformer, 48 blocks		
1	$\mathbf{m} \leftarrow \mathbf{m} + \text{RowAttention}(\mathbf{m} \mid \mathbf{z})$	node → node, edge biases the weight	Alg. 7
2	$\mathbf{m} \leftarrow \mathbf{m} + \text{ColumnAttention}(\mathbf{m})$	across the sequences at one node	Alg. 8
3	$\mathbf{m} \leftarrow \mathbf{m} + \text{MLP}(\mathbf{m})$	per node	Alg. 9
4	$\mathbf{z} \leftarrow \mathbf{z} + \text{OuterProductMean}(\mathbf{m})$	node pair → edge	Alg. 10
5	$\mathbf{z} \leftarrow \mathbf{z} + \text{TriangleMultiplication}(\mathbf{z})$, out and in	edge pair → edge	Alg. 11, 12
6	$\mathbf{z} \leftarrow \mathbf{z} + \text{TriangleAttention}(\mathbf{z})$, start and end node	edge pair → edge	Alg. 13, 14
7	$\mathbf{z} \leftarrow \mathbf{z} + \text{MLP}(\mathbf{z})$	per edge	Alg. 15
	Structure module, 8 blocks with shared weights. At the start $\mathbf{s}_u = \mathbf{m}_{1u}$ and T_u is the identity.		
8	$\mathbf{s} \leftarrow \mathbf{s} + \text{IPA}(\mathbf{s}, \mathbf{z}, T)$	node → node, values moved by $T_u^{-1} \circ T_v$	Alg. 22
9	$\mathbf{s} \leftarrow \mathbf{s} + \text{MLP}(\mathbf{s})$	per node	Alg. 20
10	$T_u \leftarrow T_u \circ \text{BackboneUpdate}(\mathbf{s}_u)$	per node, in its own frame	Alg. 23
	Recycling, 4 cycles		
11	$\mathbf{z} \leftarrow \mathbf{z} + \text{Embed}(d_{uv})$, C$\beta$ distances from the frames	frames → edges	Alg. 32

Every line is $\mathbf{h}_u = \phi(\mathbf{x}_u, \bigoplus_v \psi)$ on nodes or on edges. Information moves in lines 1, 4, 5, 6, 8 and 11.

Backup (7/14): What exactly did each ablation remove?

SI 1.13.1 to 1.13.3

Row of Fig. 4a	What the model has instead
No raw MSA	the profile and the pairwise frequencies of the MSA, in one row, no column attention
No triangles, biasing or gating	axial attention on the pairs with a relative position bias, no gating
No IPA	frames from one linear projection of the single representation, no iteration
No recycling	a single pass through the network
No end-to-end structure gradients	the structure losses stop before the Evoformer, which learns from the auxiliary losses only
No templates	no template stack, four more MSA clusters instead

Baseline: the full model without self-distillation. Three or four seeds, not re-tuned

Not additive: no IPA costs 2.3, no recycling 4.5, both together 16.3 GDT

Backup (8/14): How is confidence predicted?

SI 1.9.6 and 1.9.7, Algorithm 29

Per-Residue: pLDDT

- Linear head on single vector $\mathbf{s}_i$
- Predicts 50 discrete bins of lDDT-Cα
- Supervised via cross-entropy on self-prediction
- Reported value: expected bin center $\in [0, 100]$

Pairwise: pTM & PAE

- Linear head on pair vector $\mathbf{z}_{ij}$
- Predicts distance error bins (64 bins, 0–32 Å)
- Yields Predicted Aligned Error (PAE) matrix
- Integrates over residues to give predicted TM-score

Confidence is a learned prediction, trained directly on the network's own errors

Backup (9/14): How was it trained?

Methods, “Training regimen”, SI 1.3 and 1.11.1, Table 4

Component	Training Setup and Implementation
Datasets	PDB (< 2018, $n \approx 170$k) + 350k self-distilled Uniclust30 predictions (75% batch share)
Stage 1 (Initial)	Crops of 256 residues, 10^7 samples, $\sim$ 7 days on 128 TPUv3 cores
Stage 2 (Fine-tuning)	Crops of 384 residues, structural violation loss enabled, 1.5×10^6 samples, $\sim$ 4 days
Loss Objectives	Backbone + sidechain FAPE, torsion angles, distogram, masked MSA, pLDDT, violations

Eleven days on 128 TPU cores per model; inference requires only GPU minutes.

Backup (10/14): AlphaFold 3: further changes

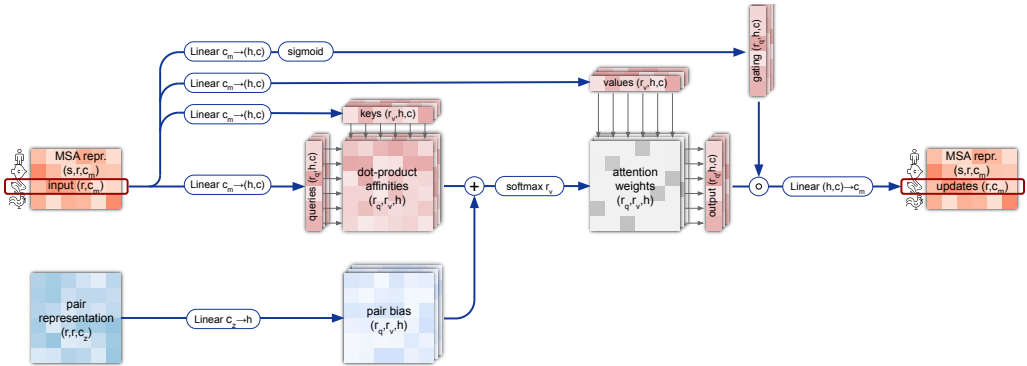
Abramson et al. [AAD+24], *Nature* 630 (2024)

Input	tokens, one per residue or per ligand atom	p. 493
Trunk	pairformer, 48 blocks. MSA block of 4, MSA not kept	p. 494
Output	diffusion over raw atoms, no frames, random rotations instead	p. 495, SI Alg. 19
Confidence	trained on 20-step mini rollouts	p. 495
Sampling	5 seeds, 5 samples each, ranked with clash and chirality penalties	p. 494, 497
Limits	chirality 4.4 %, clashes, hallucination, no dynamics	p. 496, 497

Less structure in the network, more in the data and in the sampling

Backup (11/14): MSA row attention with pair bias

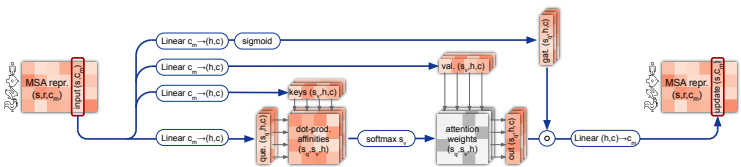
Supplementary Fig. 2, Algorithm 7



Supplementary Fig. 2. Every sequence of the MSA attends along itself, the pair representation biases the weights

Backup (12/14): Evoformer: columns and outer product mean

Supplementary Figs. 3 to 5, Algorithms 8 to 10



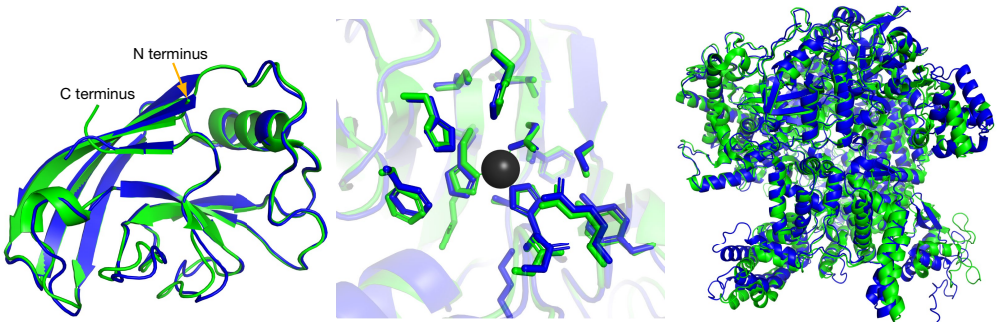
One residue at a time: the sequences attend to each other at position i, then a two-layer perceptron per cell

Outer product mean: the outer product of columns i and j, averaged over the sequences, is added to $\mathbf{z}_{ij}$

Here the alignment writes into the pairs, in every one of the 48 blocks

Backup (13/14): Prediction and experiment superposed

CASP14 targets: backbone, side chains and large chains (Fig. 1b to d)

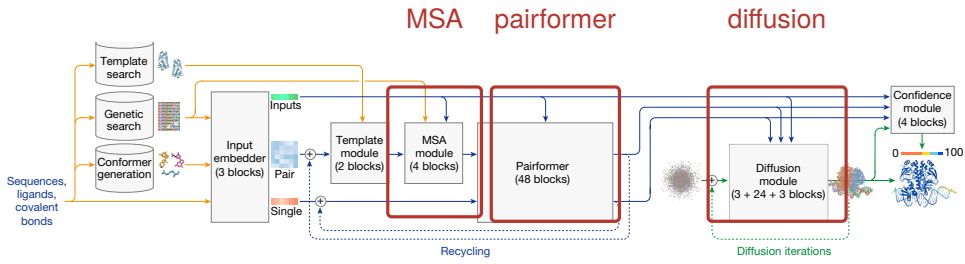


AlphaFold experiment

Prediction and experiment overlap, also in the side chains

Backup (14/14): The AlphaFold 3 architecture

Abramson et al. [AAD+24], *Nature* 630 (2024), Fig. 1d



Tokens for proteins, nucleic acids, ligands and ions

Diffusion denoises random atom coordinates, 25 samples ranked by confidence

Less structure in the network, more in the data and in the sampling